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From Data to Podium: Analytical Approaches to Understanding and Predicting Driver & Strategic Decisions in Formula 1

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Hypothesis Testing: Pit Stops during Safety Car Deployment & their Driver Position Effects.

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Abstract

This thesis explores Formula 1 pit stop strategies through advanced analytics, with a focus on driver clustering in relation to performance, tactical, and behavioural aspects. The approach led to the identification of four distinct driver categories, providing a framework to investigate pit stop strategies. The study employs linear regression as a key tool within a hypothesis-testing framework to dissect the influence of pit stops during Safety Car periods on race outcomes, especially in the context of these driver categories. This research contributes to a more refined understanding of strategic elements in Formula 1, demonstrating the role of tailored analytic methods in optimising racing tactics and decision-making processes.

Keywords

Data, Data Analytics, Sports Analytics, Formula 1, Strategy, Hypothesis Test, Linear Regression

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Common Part I

1. Introduction

In the competitive world of sports, the strategic use of analytics has revolutionised how competition is understood and won. This transformation is particularly evident in Formula 1, a sport where the smallest decisions can have significant impacts; leveraging data effectively is not merely an advantage but a cornerstone of success. Central to these strategies are the critical decisions made during pit stops, which can significantly influence race outcomes. In this context, the role of advanced analytics has become increasingly prominent, offering new avenues to optimise these crucial decisions.

While research has been conducted on various aspects of Formula 1 strategy, a notable gap exists in exploring driver clustering and its potential impact on strategic decisions as well as the evaluation of analytical approaches. This thesis is motivated by the prospect of harnessing advanced analytics and driver clustering to add additional insights to pit-stop strategies. The aim is to explore how a refined understanding of driver behaviour can enhance race outcomes and the interpretation of analytical outcomes when integrated with data-driven approaches.

This study is guided by the research question: "*How can the application of advanced analytics and driver clustering in Formula 1 enhance the accuracy and effectiveness of strategic decision-making, particularly in the context of optimising pit stop strategies?*" The objectives are twofold: firstly, to develop a robust clustering model based on performance, tactical, and behavioural metrics, and secondly, to use the results of this model within the analytical framework of a hypothesis test to assess its impact on strategic decisions.

Our methodology centres around leveraging data mainly from the FastF1 API, focusing on recent seasons to construct a clustering model, providing a multifaceted perspective on driver profiles.

The methodology extends to applying this clustering within hypothesis tests, aiming to shed light on the intricacies of pit-stop strategies and potentially add optimisations or additional insights.

This research aims to contribute significantly to the fields of sports analytics and Formula 1 strategic planning. By offering a novel perspective on driver behaviour and its implications for race strategy, the findings of this study could potentially guide teams and drivers towards more informed and effective decision-making. The insights gained could not only enrich academic discourse but also provide practical tools for strategic optimisation in the high-pressure environment of Formula 1 racing.

To lay the foundation for our analysis, we begin with a comprehensive literature review that covers the broader spectrum of sports analytics, with a specific focus on data analytics in Formula 1. This review will also discuss the application of clustering in other sports, noting its relatively nascent presence in Formula 1 literature. Following this, we detail our methodology, including our full data collection process and the clustering of Formula 1 drivers based on their performance metrics. In subsequent chapters, we examine the impact of safety cars and explore how it interacts with the identified driver behaviour clusters.

2. Literature Review

2.1. Evolution and Scope of Sports Analytics

Sports analytics refers to the application of scientific techniques for investigating and modelling sports performance. It entails organising historical data in a structured manner, applying predictive analytical models to the data, and utilising information systems to inform decision-makers (Morgulev, Azar, and Lidor 2018). This practice enables sports organisations to secure a competitive advantage through enhanced player performance analysis, game strategy formulation, health and injury prevention, fan engagement, and operational strategy optimisation (Nadikattu

2020; Tan 2023). Originating in the 1960s with notational analysis in sports like American Football and Basketball (Hughes and MFranks 2004), sports analytics has evolved significantly with technological advancements and the rise of Big Data. Today, its application extends across various sports domains, from individual sports like Tennis to team sports such as Football and has become central in the data-driven world of motorsports like Formula 1. This wide adoption underscores the broad reach and applicability of sports analytics across diverse sporting disciplines (Bai and Bai 2021).

As technology evolved, sports analytics underwent a significant transformation, unveiling new avenues for analysing and improving sports performance. A recent comprehensive review by Ghosh et al. (2023) classifies the technological advancements in sports analytics into three primary research fields: sensors, computer vision, and wireless and mobile-based applications. These areas form the bedrock of modern sports analytics, providing essential methods to collect, analyse, and interpret data. These technological advancements prove particularly pertinent in Formula 1, a sport renowned for its data-driven approach. Incorporating hundreds of sensors in racing cars facilitates real-time data collection, covering various parameters such as speed, temperature, and throttle percentage (Shapiro 2023). Effective analysis of this data offers invaluable insights for making informed decisions on pit stop strategies, car setups, and race strategies. The inclusion of artificial intelligence (AI) and machine learning (ML) algorithms amplifies the potential of these technological advancements, enabling more sophisticated analysis and predictive modelling (Ghosh et al. 2023; Dindorf et al. 2023). Integrating these novel technologies with sports analytics methodologies has opened and revolutionised research opportunities in Formula 1 racing.

2.2. Analytical Approaches in Formula 1 Racing

As established in the preceding section, sports analytics is central to enhancing competitive strategies across various sports disciplines. This becomes particularly evident in Formula 1, a sport where even the minutest decision can significantly alter race outcomes. In F1, where the stakes are high and the financial implications are vast, the synergy between data analytics and elite sporting performance exemplifies the comprehensive capabilities of sports analytics. The body of literature directly engaging with sports analytics in the context of Formula 1 is limited both in terms of the quantity of research and the range of thematic focus. Broadening the search parameters to include 'Circuit Racing Motorsports'—thereby encompassing NASCAR, Formula E, and similar series—yields an expanded body of work. Preliminary examination suggests that while there has been increasing interest in this field, it appears that the field has not yet reached a point of saturation, indicating sufficient opportunity for further research and contribution.

The existing literature can be divided into several overarching categories. However, two are predominant: lap time simulations and race simulations. As Heilmeier (2018) highlights, it is crucial to differentiate between race simulations and the more prevalent lap time simulations. The latter predominates in the literature and typically focuses on the physical or engineering aspects rather than on the holistic view of an entire race. Siegler (2000) identifies three distinct approaches to lap time simulations: Steady State, Quasi-Static, and Transient. Heilmeier (2019) published a study on quasi-static lap time simulation, applying it to both Formula 1 and Formula E to illustrate its utility. Colunga (2014) examined the modelling of transient cornering and suspension dynamics, along with the investigation of control strategies for an ideal driver within a lap time simulation framework. In a similar vein, Timings (2014) contributed to this body of work by aiming to develop a robust lap time simulation, referring to its comprehensive nature and its resilience in varying conditions.

However, for this thesis, race simulations and their components, specifically those that prioritise pit stop strategies as a key component in modelling or predicting race outcomes, are of greater relevance. Such simulations are instrumental in forecasting final standings by accounting for various factors, including driver interactions, empirical fuel consumption models, tire wear, and probabilistic effects. Bekker (2009) developed one of the earlier holistic race simulations to replicate key on-track activities in Formula 1, such as mechanical failures, overtaking manoeuvres, and pit stops. This model facilitates strategy planning by simulating the mechanical and physical dynamics of a race, thereby offering a team a potential advantage. More recently, Heilmeier (2018) outlined a simulation methodology for circuit motorsport racing strategies, which considers variables such as pit stops, tire choices, and tire degradation. The tool is designed to rapidly simulate races based on discrete lap data and adjustable strategy inputs. Building on this previous work, Heilmeier (2020) introduces a new method that further improves the simulation. This advanced version surpasses simple optimisation models by providing a comprehensive, automated simulation that responds in real-time to race dynamics. Heilmeier (2020) suggests that the current methodology for optimising pit stop decisions and associated tire compound selections might benefit from additional exploration in the future. Furthermore, he acknowledges the omission of complex strategies, such as the undercut—a tactic where a driver pits and switches to faster tires to gain time on rivals who pit later—from current models. He advocates for the integration of such tactics into subsequent models to enrich the decision-making process.

In addition to comprehensive racing simulations, focused research activities are directed at optimising specific components of the racing domain and simulations themselves, such as pit stops. These efforts aim to refine these aspects to their utmost efficiency. One research exemplifies this by employing machine learning algorithms to aid tire strategy decisions in the NASCAR series. This work utilises predictive analytics, employing historical race data to forecast positional shifts consequent to variables such as tire change frequency and tire lifespan. Extensive feature testing has revealed that support vector regression and LASSO regression yield the highest accuracy in results (Tulabandhula and Rudin 2014). Additionally, Bell (2016) advances the analysis of Formula 1 by conducting a comprehensive analysis of performance determinants in Formula 1, examining the evolving contributions of team dynamics and driver skills over time. The study results in a systematic ranking of drivers, offering insights into the qualifications of the 'potential best' based on a quantifiable set of criteria. Furthermore, Monte Carlo methods and analysis of probabilistic factors play a significant role in simulating the inherent variability present in lap times, pit stops, race incidents, and potential degradation of vehicle parts. The study of Heilmeier (2020) builds upon this foundation, providing a comparative analysis of the seminal works of Bekker (2009), Phillips (2014), and Salminen (2019), thereby extending the understanding of these stochastic elements in race simulations.

The existent body of research on Formula 1 is mainly based on the scope of publicly available data, which has been limited to lap times, and race outcomes. Such data has predominantly been sourced from the Ergast API, a privately maintained database for Formula 1 statistics (Ergast 2009). However, the introduction of the Fast F1 API marks a significant progression in data availability, offering not only the information provided by the Ergast API but also a more comprehensive set of F1 data. This includes telemetry data, official weather statistics, and track information (FastF1 2020). The introduction of the Fast F1 API thus promises to broaden the scope of current literature and models by facilitating the incorporation of mechanical details of the vehicle (such as current gear, RPM, and speed) and more granular data like positional coordinates, distances between drivers, and time specific air and track temperatures. The newly available data points, particularly telemetry data, which have received little attention in the scientific literature to date, present potential opportunities to enhance and broaden current analytical frameworks. The richer and more granular nature of this data allows for a more detailed examination of specific drivers' behaviours based on positional and telemetry information. A prospective methodological approach might include clustering drivers according to their behavioural patterns in various racing scenarios. Such clustering could yield valuable insights into performance differentiators and decision-making processes throughout a race and its strategic decisions.

2.3. Clustering Techniques in Sports Analytics and Formula 1

Clustering, a core technique in unsupervised machine learning, groups data points into distinct categories based on common attributes without predefined labels (Pedregosa et al. 2011). In sports analytics, clustering is a key tool, enabling teams and coaches to decode complex patterns and detect subtle correlations. This method can help identify performance trends and effective team compositions, which, in turn, can be leveraged to improve predictive models that forecast future sports outcomes based on the grouping of player or play characteristics. Clustering may uncover non-intuitive groupings or strategies, providing a strategic advantage in the competitive world of professional sports. Its utility and adaptability across various sports disciplines are well-documented, with a significant body of literature.

In Basketball analytics, player assessment and categorisation have evolved well beyond the confines of traditional position labels. One study by Duman, Sennaroğlu, and Tuzkaya (2021) applies hierarchical cluster analysis to a rich dataset of game-related statistics from 15 NBA seasons, uncovering four to six distinct playing styles within each traditional position. The clusters, characterised by unique attributes and skill sets, offer a multifaceted perspective on player capabilities, yielding strategic insights for player placement and team composition. Muniz and Flamand (2022) introduce an advanced clustering technique based on weighted networks. The methodology starts with k-means clustering to form preliminary groupings, which then inform a network where players are interconnected by weighted edges reflecting performance similarities.

Employing the Louvain method for community detection, the study identifies eight player archetypes, surpassing the insight offered by the five traditional positions. They further enhance this method by using tracking data, adding a layer of depth to the analysis with precise measurements of player movements and interactions. For Football, a fuzzy clustering model that can handle mixed data types has been introduced. This model assigns objective weights to attributes such as player performance metrics, positional data, and physical characteristics, thereby uncovering clusters that the complexity of mixed-attribute data might hide. Such detailed analysis facilitates the identification of player clusters according to their on-field roles, skill sets, and physical profiles, which is instrumental in tactical team structuring and player market valuation (D'Urso, De Giovanni, and Vitale 2023). A study in Tennis clustered 1188 Grand Slam players by analysing their physical and play style data, revealing four distinct profiles. Through two-step cluster analysis and further MANOVA and discriminant analysis, the research identified how factors like height and handedness correlated with performance, notably in serving and net play (Cui et al. 2019). Clustering in sports analytics reaches beyond mainstream sports, extending its application to disciplines like Badminton. Sinadia and Murwantara (2022) apply k-means and hierarchical agglomerative clustering to analyse Badminton athletes' performances, identifying clusters based on game results and consecutive scoring. K-Means clustering discerns four distinct performance clusters, with one particularly strong cluster associated with high scores and consecutive points, supported by similar results from hierarchical clustering.

Collectively, these studies showcase the breadth and depth that clustering techniques bring to sports analytics. They enable a sophisticated segmentation of athletes, offering a granular understanding that can guide coaching decisions, athlete development, and competitive strategy formulation. In the data-rich context of Formula 1, the application of clustering techniques to group drivers by various data-driven similarities could potentially yield insights that extend beyond traditional performance metrics. While such methods have provided detailed understandings in other sports disciplines, the extent to which they have been applied to driver analysis in Formula 1 remains limited. Most existing studies on driver performance within this sport have focused on race results without a significant exploration of specific clustering techniques that have been beneficial in broader sports analytics.

For instance, one pioneering study by Phillips (2014) presents a statistical model that quantifies the contributions of drivers and teams to performance, using championship points as the metric. In another notable work by Rockerbie and Easton (2021), they employ an econometric method to dissect the relative impact of driver skill versus car technology on race results, discussing the "80-20 rule" but primarily concentrating on race positions and outcomes without delving into driver-specific skill and performance metrics. Similarly, a recent study utilises a Bayesian multilevel rank-ordered logit model to analyse historical ranking data, aiming to distinguish between driver skill and constructor efficiency (Van Kesteren and Bergkamp 2023). However, like its predecessors, it does not extensively investigate drivers' characteristics or behaviour.

While providing valuable insights, the discussed studies often centre around broader performance metrics and race outcomes without specifically exploring the clustering of drivers based on their attributes or behaviours. This gap indicates an opportunity for further research into the application of driver clustering techniques within Formula 1. Exploring driver clustering, especially by using detailed telemetry data, expands the analytical horizon and could establish a foundation for in-depth examination of strategic components, such as pit stop tactics and other decision-making processes in the sport. Furthermore, the comprehensive use of telemetry data remains underutilised in existing models. Incorporating telemetry data into clustering analyses could reveal more profound insights into driver behaviour, driving styles, and performance metrics, which go beyond the conventional race outcomes and rankings.

3. Data

3.1. F1DataFetcher

The publicly accessible API from FastF1 and the associated Python library can be used to generate the required data. In contrast to the previously common ERGAST API, the FastF1 API also includes interesting aspects such as weather, position and telemetry information in addition to the usual Formula 1 data. Telemetry information includes speed, revolutions per minute and much more. Position data not only informs about the exact position of a car on the racetrack but also provides information about direct duels by showing which driver is driving in front of a driver and how far away he is. This data can be a great asset and a clear advantage over existing literature. That's why the decision on FastF1 as the main source of data was made quickly. Using this API is relatively simple and works without any problems. Interested readers are referred to the detailed documentation provided by the operators (FastF1 3.1.6). However, the API has one limitation for efficient data collecting purposes: the user can only ever load one event of a session, be it qualifying, race, free practice, etc., from the API. This is not due to the user's authorisation but merely to the logic of the API. To get around this, a new logic was created.

The Python class we developed is called F1DataFetcher. This allows us to pull data for multiple races simultaneously from the API into a notebook, automatically creates the required data frames in the required structure and outputs them as Panda's data frames. In addition to the challenge of being able to receive multiple data at once, it was also important to be able to save the data so that local access without the need of reloading data every time was granted. Therefore, we created a relational database in PostgreSQL. This allows to save the individual data frames that are collected by the API via the F1DataFetcher in a meaningful, coherent, and comprehensible way and to keep them ready for further use. The database was created manually in pgAdmin4, and a connection was established via the Python library ‘sqlalchemy’. This package allows users to send SQL commands in Python to SQL-based databases and receive pandas’ data frames. The methods of the FastF1 Data Fetcher were enhanced by the capability of sending collected data automatically to a specified PostgreSQL database. The final relational database contains a total of 10 tables. The structure and relations are composed as shown in Figure 1.

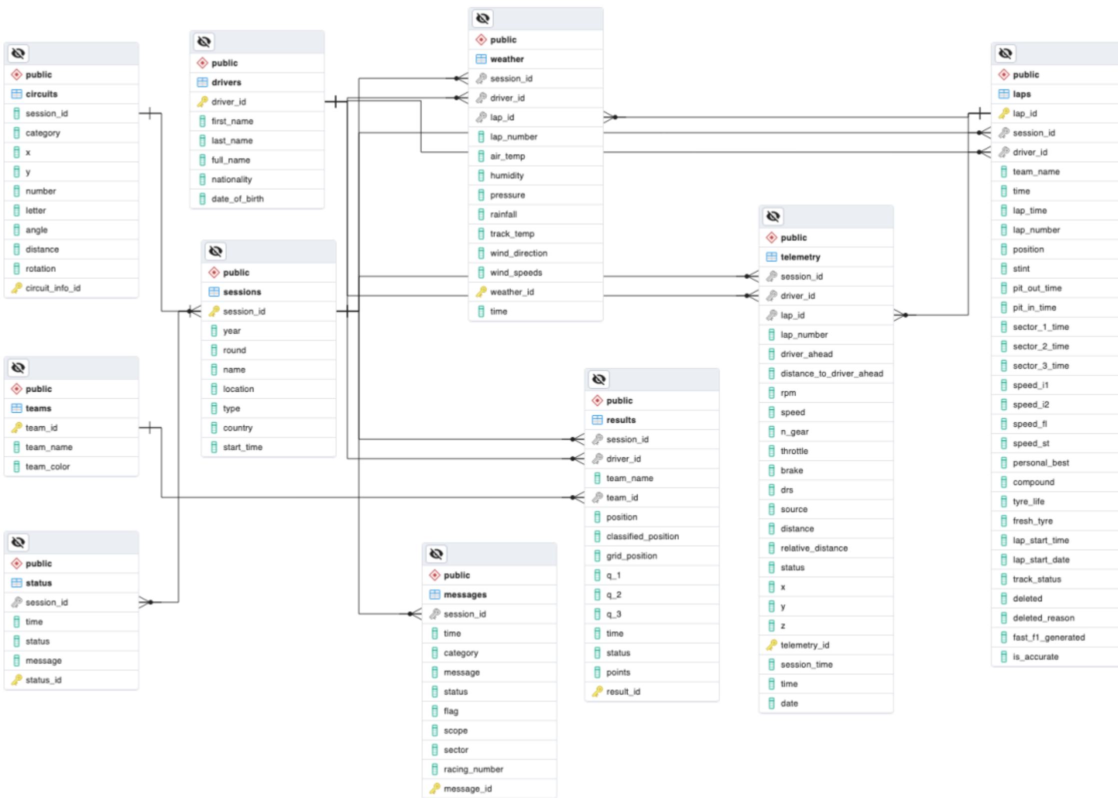

Figure 1: Simplified Schema of Our Own Data Base

The final F1DataFetcher class therefore contains ten methods plus one more to collect different data frames from FastF1 (and ERGAST API) and optionally store them in a database with the above schema. The data frames are on different levels which will be explained in the following along with the methods used to collect this data.

The first group of methods collect general information, such as drivers or teams. Although this data is collected at the session level, the information is overarching and not tied to individual sessions. A comprehensive list of the methods used to collect this data is shown in Table 1.

Table 1: F1DataFetcher Methods - Group 1

Method	Description
get_multiple_sessions()	Load detailed information on grand prix events, optionally store to database
get_unique_drivers()	Load detailed information on unique drivers, optionally store to database
get_unique_teams()	Load detailed information on unique teams, optionally store to database

The second group of data is the one where data is on session level. The methods that collect these kinds of data are listed in Table 2.

Table 2: F1DataFetcher Methods - Group 2

Method	Description
get_session_results()	Load detailed information on grand prix results, optionally store to database
get_race_control_messages()	Load detailed information on race control messages, optionally store to database
get_track_status()	Load detailed information on flags and statuses, optionally store to database
get_circuit_info()	Load detailed information on track characteristics, optionally store to database
get_weather()	Load detailed information on weather characteristics, optionally store to database

The final set of methods collect data that is highly granular, meaning collected individually on lap basis for each driver. A full overview is provided in Table 3.

Table 3: F1DataFetcher Methods - Group 3

Method	**Description**
get_laps()	Load detailed information on laps, optionally store to database
get_telemetry()	Load detailed information on telemetry, highly granular
store_to_postgresql()	store highly granular telemetry data to database

The methods of the F1DataFetcher provide a complete, easy to use way of collecting large amounts of data from FastF1 API and automatically store in PostgreSQL. They were then applied to set up a full database that served as the input data for the complete work.

3.2. Data Limitations

For feature engineering, data pre-processing and normalisation, data was limited for better performance and other reasons. The process of data filtering was conducted with the following steps:

- **Exclusion of races during rain**: Grand Prix events in rainy weather conditions were excluded. The reason behind this was that in the complete database, there is a lack of data for events in the rain. Since not all drivers participated in all the races, and metrics can be significantly different on wet tracks, the decision was made to eliminate these data points from the data that serves as input for feature engineering.
- **Removal from retired drivers**: Although there is some information in the data about why they retired, there is no sufficient way to know if it was their fault, possibly some other driver's fault, a tactical decision or simply a problem with the car. Therefore, the decision was made to remove all data of drivers who retired on a particular track and only look at those data points where a driver successfully drove all laps of the race.

- **Removal of interrupted laps**: Sometimes races get interrupted due to accidents, weather conditions or other reasons. In these cases, the data is noisy, leading to lap- and pit stop times of several hours, which makes no sense and only distorts some of our engineered features. To avoid these kinds of complications, the data for laps in which a red flag was hissed, meaning the race was interrupted, was omitted from the data.

4. Clustering

4.1. Introduction to Clustering in Racing Analysis

Transitioning to clustering, a principal method in unsupervised machine learning, we leverage the refined dataset to explore and interpret the complex dynamics of Formula 1 racing.

Clustering is essential in the analysis and interpretation of complex data sets. This technique, as elucidated by Jain (2010), involves the organisation of a collection of patterns into clusters, based on similarity. Patterns within a cluster exhibit a higher degree of similarity to each other than to those in different clusters, thereby revealing the intrinsic structure of the data. The significance of clustering extends beyond mere data categorisation. Jain (2010) emphasises its role in exploratory data analysis, summarisation, and compression, as well as its utility as a tool for hypothesis generation and testing. In diverse fields such as image and pattern recognition, bioinformatics, and information retrieval, clustering is instrumental in uncovering hidden structures, identifying anomalies, and simplifying complex data landscapes.

In the realm of sports analytics, particularly in Formula 1 racing, the application of clustering takes on a more defined and critical role. In the high-octane world of Formula 1 racing, understanding the nuances of driver behaviour is essential for gaining a competitive edge. This study employs a clustering methodology to delve into the intricacies of driving behaviour, aiming to achieve a set of specific objectives that enhance our comprehension of what differentiates top performers on the track. A central goal of this research is the characterisation of driver profiles through clustering. This involves categorising drivers into distinct groups based on a comprehensive array of performance metrics and observed behaviours on the track. The aim is to construct detailed profiles that capture the essence of each driver's approach to racing, including their performance, behaviour, and tactics. These profiles are expected to provide a holistic view of each driver's strengths and areas for improvement, offering insights into the diverse strategies and techniques employed in Formula 1 racing. Furthermore, the study engages in a comparative analysis across the derived clusters. This analysis is crucial for identifying the unique characteristics that distinguish each cluster, thereby shedding light on what sets apart the most successful drivers. This comparative approach is anticipated to reveal valuable insights into the factors that contribute to effective driving strategies and overall race performance.

Finally, the clustering analysis aims to provide practical strategic implications for teams and drivers. By understanding the distinct behavioural clusters, teams can tailor their training programs, strategy development, and in-race decision-making to better align with the identified strengths and weaknesses of their drivers. This objective is predicated on the belief that data-driven, personalised strategies can significantly enhance performance and competitiveness in Formula 1 racing. Aligning training and strategies with the insights gained from clustering analysis, teams can optimise their approach to each race, maximising their potential for success in this dynamic and challenging sport.

4.2. Feature Engineering

Before being able to cluster driver behaviour and performance, it is vital to define metrics and input features that capture the essence of what is tried to analyse. Since Formula 1 is a complex competition with various influential factors, we decided to limit it down to three main aspects: The performance metrics employed to evaluate a driver's performance, behavioural metrics that illustrate drivers' conduct during driving, and tactical metrics, reflecting the strategic choices made by a driver or team throughout a race.

In the following section, an overview of the features created for the clustering of Formula 1 drivers is represented, detailing the data utilized for their calculation.

4.2.1. Performance Metrics

Despite the significant dependency of performance on the car's capabilities, for which we have insufficient data, our objective was to quantify specific dimensions of driver performance. We defined three main aspects of driver performance: lap time, position gain, and consistency in driving. The first two metrics are self-explanatory and do not require additional clarification. The third metric, consistency, aims to quantify a driver's uniformity on the track, specifically regarding lap time variations. These three metrics effectively represent a comprehensive assessment of a driver's performance.

Table 4: Clustering - Performance Metrics

Variable	Name	Description
Lap time	avg_lap_time	Average lap time for each driver on all races
Position gain	position_gain	Average number of positions gained (or lost)
Consistency	consistency	Standard deviation in average lap time

4.2.2. Behavioural Metrics

More complicated than defining metrics to see a driver's performance is mapping various aspects of driving behaviour to input features. The focus was primarily on aggressiveness and a driver's behaviour towards other drivers, especially the driver ahead of him. To analyse behaviour in direct competitive scenarios, we quantified various gaps and distances relative to the leading driver, thereby attempting to delineate patterns of conduct in these head-to-head duels. After all, the five new input features for capturing driver behaviour were defined as the *distance to the driver ahead*, the *time within range*, *persistence*, *overtakes*, and *defensive actions*.

Table 5: Clustering - Behavioural Metrics

Variable	Name	Description
Distance to driver ahead	avg_distance	Average distance to the opponent ahead
Time within range	time_within_range	Average time a driver spends in a range of 50 meters behind his opponent
Persistence	persistence	Standard deviation of the distance to the driver ahead
Overtakes	overtakes	Average number of successful overtakes
Defensive actions	persistence	Average number of laps without losing a position

4.2.3. Tactical Metrics

As tactical decisions that are made by team leads and the driver himself during constant communication play an essential role in Formula 1 and can impact a driver's final position and overall performance significantly, capturing differences in strategic decisions was the goal that was aimed for. Two main aspects of Formula 1 tactics are pit stop timing and compound choice. Deciding when to pit stop, which tires to wear and how many pit stops a driver does can drastically change the outcome of a race. To capture the strategic decisions for each driver, a new set of features was created from the data to capture the pit stop timing and the compound strategies.

Table 6: Clustering - Tactical Metrics

Variable	Name	Description
Pit time	avg_distance	Average time a pit stop takes
Pit window: Early	early	Percentage of pit stops in first 25% of the laps
Pit window: Mid	mid	Percentage of pit stops in mid 50% of the laps
Pit window: Late	late	Percentage of pit stops in last 25% of the laps
Laps on SOFT	laps_on_soft_percentage	Percentage of laps driven on SOFT compound
Laps on MEDIUM	laps_on_medium_percentage	Percentage of laps driven on MEDIUM compound
Laps on HARD	laps_on_hard_percentage	Percentage of laps driven on HARD compound

4.2.4. Normalisation

Normalisation is a common technique to scale data to a standard range, eliminate redundant or noisy data, and improve algorithm performance significantly (Patel and Mehta 2011). Literature shows that it is common practice in machine learning to apply normalisation techniques to data before using it as input data (Gopal, Patro, and Kumar Sahu 2015).

Since most clustering algorithms, such as K-means and others, rely on scaled data, we needed to normalise our data points. Another reason to follow that approach is that in motorsports in general and Formula 1 in particular, the information on the actual car build is highly confidential and, due to this, not publicly accessible. The cars have different characteristics not shared by the teams participating in Formula 1. There might be cars performing better in general than other cars. Since the focus was merely on the driving capabilities of the individual drivers only, a need to reduce the potential bias that could influence our clustering algorithms was urged.

Besides that, some input features can be significantly influenced by track characteristics, such as the number of corners on a track and their angle. Since data from various Grand Prixes was used, results achieved on multiple tracks were compared. To make them comparable, again, there is a need for data normalization.

There are several ways of scaling and normalising data, the most common being z-score scaling and min-max normalisation. For this purpose, the decision was made to apply the latter to our input data. Min-max normalisation is a technique that keeps the original relationship among the data (Gopal, Patro, and Kumar Sahu 2015). It follows a simple calculation:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

The Python library 'scikit-learn' (Pedregosa et al. 2011) provides a standard pre-processing algorithm for this kind of normalisation that we applied to our data.

4.3. Selection of Clustering Algorithm

After normalising our data to ensure comparability and reduce potential biases, as detailed previously, we move to the next crucial step in our analysis: the selection of an appropriate clustering algorithm. The choice of an algorithm is crucial, as it must align with the nature of our normalised data and the specific objectives of our study. The selection of the K-Means clustering algorithm for the analysis of driver behaviour in Formula 1 racing is underpinned by several compelling reasons that align with the specific nature of the data involved. A primary factor is the simplicity and computational efficiency of K-Means, as highlighted in the study of Jain (2010). This characteristic is particularly advantageous given the large volume and complexity of data generated in Formula 1. K-Means offers an efficient and straightforward approach, which is essential for the initial exploratory analysis where a fundamental understanding of data grouping is sought. Additionally, the suitability of K-Means for clustering numerical data, as noted by Arthur and Vassilvitskii (2007), aligns well with the mostly numerical nature of our engineered data. This alignment ensures that K-Means is well-equipped to handle the specific types of data encountered in this research. In addition, the efficiency of K-Means in processing large datasets is a significant advantage, as it allows for the swift and effective handling of the extensive data typical in Formula 1 racing. This efficiency, coupled with the ease of interpretation of its results, as Jain (2010) notes, makes K-Means a practical choice for researchers and analysts who require clear and actionable insights from their data.

However, it must also be noted that K-Means has certain limitations in its application. One challenge is that it operates under specific assumptions about the nature of the clusters it forms. Typically, the algorithm assumes that clusters are spherical and of similar size. However, as Arthur and Vassilvitskii (2007) point out, these assumptions may not always hold true in real-world data scenarios, including those encountered in Formula 1 racing. This limitation necessitates an understanding that the clusters formed may not perfectly represent the complex and varied nature of the data.

Additionally, K-Means is known to be sensitive to the initial choice of centroids. This sensitivity can lead to variability in results across different runs of the algorithm, as observed by Celebi, Kingravi, and Vela (2013). This characteristic requires careful consideration and potentially multiple iterations to ensure robustness in the clustering outcomes. This need for a precise selection of starting points leads directly to the next crucial step of our analysis: determining the optimal number of clusters.

4.4. Algorithm Parameters and Tuning - Determining the Number of Clusters

In defining the number of clusters for our k-means analysis, we employed a diverse approach to ensure methodological rigour. Initially, we utilised the elbow method, widely recognised for its effectiveness in cluster analysis (Kodinariya and Makwana, 2013). This method involved plotting the explained variance against the number of clusters and identifying an 'elbow' point where the rate of decrease in variance sharply changes. Our analysis indicated an elbow point between three and four clusters, suggesting that further increasing the number of clusters would result in marginal improvements in explained variance.

To complement this quantitative method, domain knowledge regarding the performance of drivers was also incorporated. This additional layer of analysis provided valuable context in guiding the decision-making process. Specifically, the decision to opt for four clusters was influenced by the distinct characteristics of drivers Russel and Latifi, who formed a separate fourth cluster in contrast to the other three clusters. Given their different tire choices compared to the other drivers and their overall race positioning at the lower end of the grid, it was deemed appropriate to assign them to a separate cluster. This decision underscores the importance of integrating quantitative methods with domain-specific knowledge for more insightful and informed analyses in complex situations.

While the silhouette score, a measure of cluster cohesion and separation, was also calculated as part of our evaluation, it was the combination of all three together that ultimately led us to choose four clusters. This number of clusters appeared to offer a reasonable balance, ensuring that the clusters were distinct and meaningful without introducing overfitting or unnecessary complexity. The decision to opt for four clusters was made with the goal of achieving a segmentation that would allow for an insightful analysis of driver behaviour.

4.5. Results

Applying the outlined clustering methodology, drivers in the Formula 1 season of 2019, 2020, and 2021 were effectively segregated into four distinct clusters, which can be seen in the scatterplot in Figure 2. This is based on the presented combination of performance, tactical, and behavioural metrics.

The allocation of drivers across these clusters is as follows:

- **Cluster 0:** Albon, Raikkönen, Perez, and Magnussen.
- **Cluster 1:** Hamilton, Verstappen, and Bottas.
- **Cluster 2:** Grosjean, Kvyat, Leclerc, Norris, Ocon, Gasly, Ricciardo, Sainz, Stroll, Giovinazzi, and Vettel.
- **Cluster 3:** Latifi and Russell.

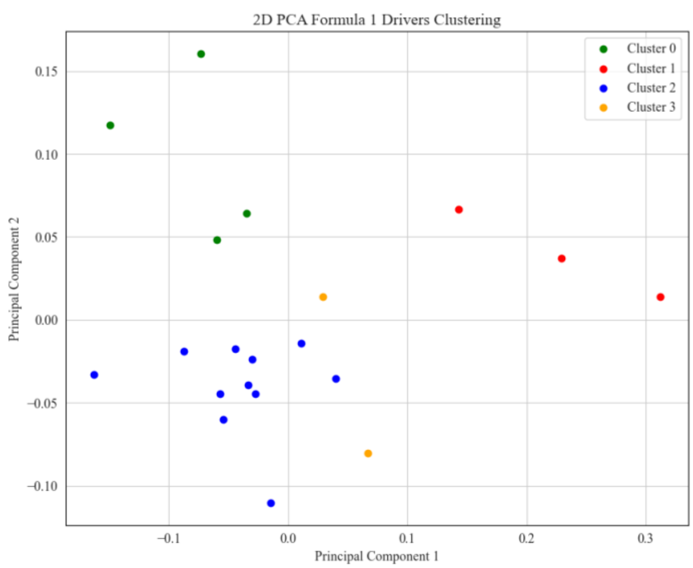

Figure 2: Scatterplot Illustrating Driver Clusters in 2D PCA Space

An initial analysis reveals a reasonable distribution reflective of known performance tiers within the sport. The top performers, typically dominating the front of the pack, are distinctly grouped in Cluster 1, while those frequently at the rear form Cluster 3. The midfield drivers, recognised through domain knowledge as a diverse group in terms of tactics and performance, are split between Clusters 0 and 2.

To describe the clusters in more detail and to interpret the classification, the underlying performance, tactical and behavioural metrics, which were defined as input features, need to be put into perspective. The different degrees of these features can also be seen in Figure 3. Here, the mean of the features per cluster is shown in the form of a parallel plot, enabling a visual comparison between the clusters.

4.6. Interpretation

When interpreting the clustering results, it is important to recall that in Formula 1, the competition involves the top 20 drivers in the world. This elite group ensures a high degree of skill parity, as illustrated by our parallel plot analysis. The analysis underscores the close competition and high level of competence inherent in this sport. However, upon closer examination, subtle but meaningful variations in performance can still be observed. It's crucial to emphasise that while these differences are partly due to the varying budgets of the teams and, consequently, the performance of the cars, the vehicle is not the sole determinant of a driver's overall performance. Teams with larger financial resources often have an advantage in terms of car performance, but the skills and decisions of the driver are also key to success. Teams with limited budgets may face challenges in vehicle performance, yet the individual performance and strategy of the driver can mitigate these disadvantages to some extent. This interplay between car performance and driver skill is a key element in analysing the competitive dynamics of Formula 1.

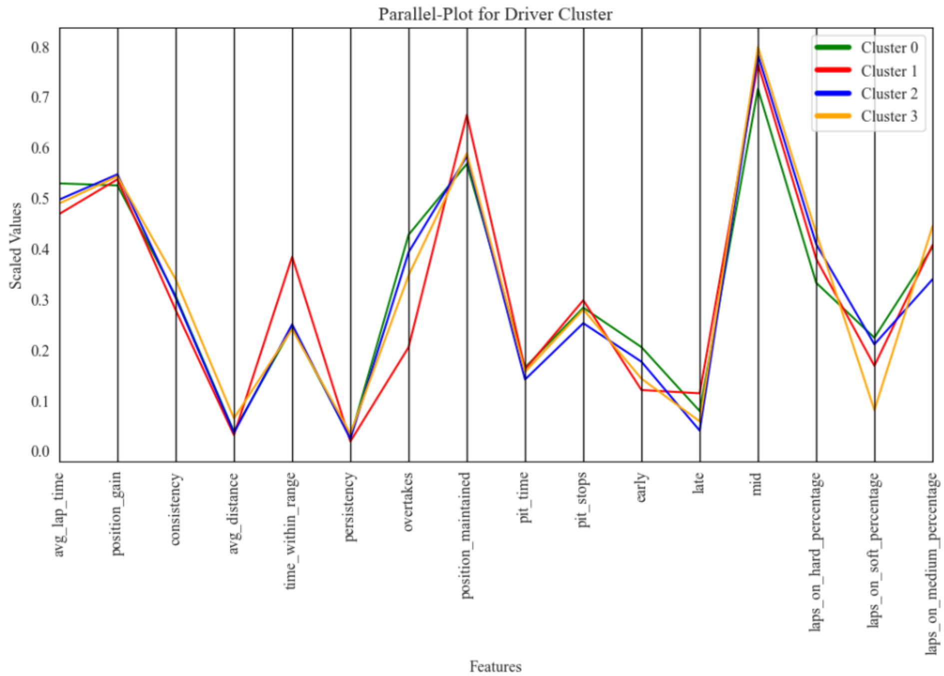

Figure 3: Parallel Plot Showing Mean Features of Driver Cluster

In our subsequent analysis, we will explore the specific characteristics of the clusters, bearing in mind the influence of vehicle differences, to provide a comprehensive understanding of the factors driving performance in Formula 1. This detailed examination starts with Cluster 0 and is based on the mean of each feature for each cluster.

Drivers in Cluster 0 are characterised by above-average lap times, suggesting a focus on endurance and tactical positioning rather than outright speed. Their notable position gains reflect effective racing behaviour, with a high number of overtakes suggesting skill in dynamic racing scenarios. The moderate consistency of their performance can potentially be attributed to the positioning of the drivers in the midfield. Interactions between the drivers are much more frequent here, which has a greater impact on lap times. These interactions are also reflected in the frequent overtaking manoeuvres. The cluster's pit strategy, marked by average pit times and a preference for early stops, points to a tactical edge in race management. This is complemented by their significant usage of hard tires, implying a strategy centred on durability and long-term race planning over the immediate speed offered by soft tires.

In essence, drivers in Cluster 0 excel in strategic manoeuvring and consistency, prioritising race longevity and tactical positioning over peak lap speed. This approach aligns with drivers who may not have the fastest cars but utilise strategic race management to optimise their overall standings.

Moving on to Cluster 1, which is characterised by the lowest average lap times, indicating superior performance. Their excellent position maintenance skills, combined with the highest average amount of pit stops, point to strategic control of both race pace and pit strategy. The frequent pit stops in this strategy are likely a tactical decision to optimise tire performance, considering there is usually sufficient time for an additional pit stop towards the end of the race without losing positions. This approach aligns with the balanced tire usage observed, where a high percentage of laps are run on both medium and hard compounds. Their exceptional consistency underscores a notable ability to deliver stable, reliable performances across various races. Additionally, their highest time within range of the driver in front, paired with a lower number of overtakes, suggests they often lead races and, if not remain close to the driver in front, efficiently managing their positions and indicating an aggressive approach to overtaking manoeuvres. In essence, Cluster 1's drivers demonstrate a blend of speed, strategic expertise, and consistent performance, marking them as elite competitors in the Formula 1 field.

Next is Cluster 2, which presents a profile of competent race management with a focus on strategic tire usage. This cluster's average lap times and position gains, along with a consistency level akin to Cluster 0, indicate a balanced approach to race performance, matching speed with strategic positioning. The drivers in this cluster exhibit a lower frequency of overtakes compared to Cluster

0, yet they maintain their positions effectively during races. This suggests a focus on consistent lap performance and strategic placement rather than aggressive overtaking manoeuvres. Their ability to hold positions is indicative of skilful ability, particularly in defending against competitors and capitalising on track opportunities. A notable aspect of Cluster 2 is their pit strategy. These drivers tend to pit the earliest among all clusters and have the lowest average number of pit stops. The fewer pit stops suggest a preference for longer stints on track. This aligns with their unique tire usage pattern, as Cluster 2 is the only group to perform more laps on hard tires compared to medium ones. This preference indicates a strategy leaning towards tire durability and longer race stints, possibly to maintain consistent lap times and reduce the time lost in pit stops. The hard tire's longevity would be particularly advantageous in races with high tire degradation or where preserving tire life is crucial for race strategy.

In summary, Cluster 2 drivers excel in strategic racing behaviour and tire management, combining average lap times with effective position holding and a distinctive early pitting strategy, highlighting a focus on endurance and tactical adaptability in Formula 1.

Lastly, Cluster 3 presents a unique set of characteristics. Their lap times and position gains are comparable to those in Cluster 2, suggesting a similar approach in terms of speed and race advancement. However, the distinctive feature of this cluster is the lowest consistency among all groups, as indicated by the highest standard deviation in average lap times. This variability could be attributed to a range of factors, such as adapting to different race conditions, varying strategic choices, or fluctuations in both driver and car performance. Another notable aspect of Cluster 3 is their racing style, which seems to focus on endurance and stability. This is inferred from their longest average distance to the driver ahead. Such characteristics may imply a tendency towards a more calculated and defensive racing approach, possibly prioritising maintaining a steady pace and avoiding risks. The tire usage pattern in this cluster further supports this interpretation. The drivers in Cluster 3 have the lowest percentage of laps on soft tires, which are typically used for aggressive, short-term speed advantages. Their reluctance to use soft tires might indicate a preference for strategies that emphasise tire longevity and sustained performance over short bursts of speed. This approach is often seen in races where managing tire degradation and fuel consumption is critical for achieving a favourable outcome.

In conclusion, Cluster 3's drivers appear to adopt a strategic approach focused on endurance and stability, with less emphasis on aggressive manoeuvres and high-speed performance. Their lower consistency might reflect a more adaptive or variable strategy, responding to the specific demands of each race. This approach, combined with a cautious tire strategy, suggests a focus on long-term race management rather than immediate position gains.

In summary, the clustering analysis of Formula 1 drivers reveals distinct strategic profiles across the four groups. Cluster 0 and 2 both emphasise endurance and tactical positioning, but Cluster 2 distinguishes itself with an earlier pitting strategy and a greater emphasis on hard tires. Cluster 1 stands out for its combination of high speed, strategic pit stops, and consistent performance, showcasing the traits of front-running drivers. In contrast, Cluster 3 is defined by its variability in performance and a more conservative tire strategy, indicating a focus on stability and endurance. This comparative analysis highlights the diverse approaches in Formula 1, from aggressive speed and tactical prowess in Cluster 1 to the more calculated and endurance-focused strategies in Clusters 0, 2, and 3.

Individual Part

5. Hypothesis Testing: Pit Stops during Safety Car Deployment & their Driver Position Effects

5.1. Introduction

Formula 1 racing is characterised by the potential for crashes and race disruptions, requiring strategic adaptations to these high-stakes situations. A pivotal element in managing these incidents is the deployment of the Safety Car (SC) and Virtual Safety Car (VSC), which significantly alters the teams' and drivers' dynamics and strategies. The SC is typically deployed during dangerous incidents, such as accidents or hazardous conditions, to ensure safety by leading the drivers at a reduced speed. This results in the “neutralisation” of the race, compressing gaps between cars and prohibiting overtaking. The introduction of the SC creates a strategic problem where teams must swiftly decide whether to call their drivers in for a pit stop, considering factors like car position, remaining race distance, and tire wear. A well-timed pit stop during SC can minimise time lost, whereas staying out may lead to pitting under green flag conditions, potentially losing more time. Similarly, the VSC, activated for less severe incidents, imposes a speed limit on the track, presenting similar strategic implications to the SC, albeit often to a lesser extent.

This part of the thesis focuses on investigating how pitting strategies during SC and VSC deployments influence a driver’s position in the race. The subordinated research question is:

"*Does pitting during an SC or VSC deployment significantly influence a driver's net gain or loss in finishing positions compared to those who do not pit under the same conditions?*"

To explore this, we propose the null hypothesis (H0), which suggests that pitting during an SC or VSC deployment does not significantly affect a driver's finishing position gain or loss. The alternative hypothesis (H1) counters this by suggesting a significant advantage or disadvantage for drivers who pit during these periods. To test these hypotheses, we'll conduct an exploratory data analysis (EDA) and develop a multiple linear regression model. Additionally, further regression analysis based on driver clusters will compare results to our primary findings, enriching our understanding of the Formula 1 pit stop strategy. The aim is to provide a detailed insight into the strategic decision-making during SC periods in Formula 1, blending theoretical analysis with practical implications for teams and drivers.

5.2. Data & Methodology

5.2.1. Data Pre-Processing

For this analysis, we utilised data from the 2019 to 2021 Formula 1 racing seasons, a period selected for its alignment with the previous driver clustering as outlined in the limitations section. The resulting dataset encompasses 57 races, offering a substantial sample size for robust statistical analysis. Key data tables relevant to the research question include *Results, Status, Laps* and *Weather*. A critical step in ensuring the accuracy and relevance of our data involves a further cleaning process suited to this specific research.

This involved removing atypical races, like the 2021 Belgian Grand Prix, which was cancelled after three laps due to heavy rain, offering limited scope for strategic decisions. Similarly, the 2019 German Grand Prix was excluded due to its abnormal conditions, leading to multiple safety cars and pit stops, which were outliers in our analysis. These exclusions were crucial for maintaining the consistency and integrity of the dataset, ensuring it accurately reflected standard race dynamics for a valid analysis. Moreover, data from drivers who did not finish (DNF) their races, along with their associated laps, pit stops, and results, were also excluded. This decision is crucial because DNF drivers lack a final race position, which is essential for the analysis centred on assessing position changes. We specifically examine the influence of pit stop strategies on finishing positions, making the inclusion of data from drivers without a complete race outcome irrelevant to this investigation.

5.2.2. Feature Engineering & Feature Selection

The next crucial step in our analysis is engineering features based on our existing dataset. Feature engineering is instrumental in transforming raw data into meaningful variables that effectively represent the dependent and independent variables crucial for hypothesis testing. The dependent variable to explain in this study is the *position_change_score*, a metric designed to quantify a driver's relative performance regarding position gain or loss. Instead of using only the net position change, which may not fully capture the nuances of race dynamics, we developed this score to provide a more comprehensive performance assessment. It is calculated as follows:

- *position_change_score*: The score is the weighted change in a driver's finishing position relative to their position immediately after the SC/VSC period. The weights assigned decrease incrementally for lower positions, starting with a weight of 1.4 for a change to 1st place, 1.35 for 2nd place, 1.3 for 3rd place, and so forth, tapering down to a default weight of 1.0 for the rest of the positions.

The rationale behind this weighting system is the assumption that position changes at different levels of the race grid are not equally challenging or strategically significant. A change in position towards the front is often more difficult to achieve and holds greater strategic value, hence the higher weights.

To further refine our analysis, we categorise the variables into an independent variable and several control variables. Each variable is selected for its potential impact on the dependent *position_change_score*. The independent variable is central to the analysis and hypothesis testing, focusing on a critical strategic decision during the race:

- *pitted_during_sc_vsc*: A binary variable that indicates if a driver has made a pit stop during an SC or VSC period.

The control variables are included to consider other factors that could affect the score, thereby capturing the multi-faceted nature of a race:

- *sc_type*: A binary variable differentiating between a physical SC and a VSC.
- *pitstop_count_before_sc_vsc*: The number of all pit stops a driver has made before the SC/VSC period.
- *all_pitstops_late*: The percentage of pit stops in the last third of the race compared to all pit stops.
- *pitstop_time_avg_seconds*: The average duration of all pit stops of a driver during the race.
- *tire_change*: A binary variable indicating whether a tire change occurred during a pit stop under SC/VSC conditions.
- *tire_life*: The variable quantifies all laps completed on the current tire compound throughout the entire circuit.
- *position_change_after_sc_vsc*: The variable measures the immediate shift in a driver's position after the end of the SC/VSC period.
- Other factors that are not pit stop related: *start_position, overtakes* (successful overtakes per driver), *team_name, rainfall.*

Incorporating these variables allows for a comprehensive analysis of the factors influencing the dependent variable, particularly in understanding how SC/VSC-related pit stops affect driver outcomes in a race. To accurately analyse these factors in our regression model, we methodically addressed the categorical variables through targeted transformations. Binary variables, such as *sc_type*, underwent binary transformations to create clear indicators for specific conditions like SC or VSC. This method ensures these binary states are adequately represented in the analysis. For multi-category variables like *team_name*, dummy coding was utilised. Each Formula 1 team was represented by its binary column, with *Red Bull Racing* chosen as the reference category for comparison. Furthermore, a log transformation was applied to counter potential skewness in the distribution of the dependent variable. This step is critical for normalising the data, aligning with the normality assumptions required for linear regression.

5.2.3. Exploratory Data Analysis

The EDA focused on the strategic aspects of SC and VSC deployments, particularly examining SC/VSC pit stops and their influence on the *position_change_score*. Recent seasons reveal a consistent pattern showing that traditional SCs are deployed more frequently than VSCs, as depicted in Figure A.1 in Appendix A. This recurring prevalence of SC deployments sets an important context for the analysis, as the more frequent activation of the SC may suggest a greater number of strategic opportunities for teams to make pit stops compared to VSC periods. Further extending this analysis to the duration aspect, we find that SC periods typically last longer than VSCs (see Fig. 4). This difference in duration is not merely a procedural detail but carries significant strategic weight. The extended time SCs spend on the track implies more substantial windows for strategic pit stop decisions. In contrast, the shorter VSC durations may limit the feasibility and attractiveness of pit stops, leading to most strategic pit stop activities occurring

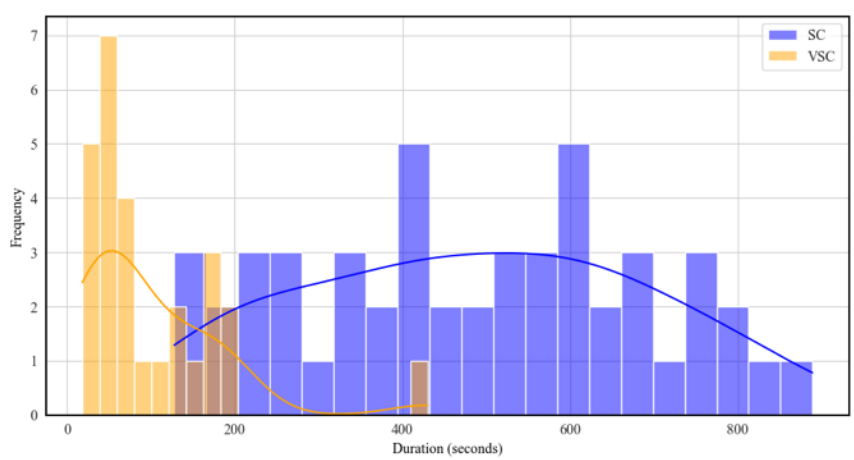

Figure 4: Comparison of durations of SC and VSC pitstops

during the longer SC periods. An interesting pattern emerges when looking at the pit stop frequencies in Figure A.3 in Appendix A,

indicating that teams tend to favour executing pit stops primarily in the early to mid-stages of races during SC periods. This trend suggests a deliberate team strategy to utilise the SC conditions, which isn't solely about the benefit of a slower pace. Pitting during these critical moments allows teams to adapt to the evolving race while fully leveraging the advantage of fresh tires. Additionally, executing pit stops early in the SC period helps avoid the congestion and tactical complexities that may arise later as the race pace resumes. In contrast, VSC periods, typically shorter in duration, may not align with such strategic opportunities, leading to differing approaches to pit stops under these conditions.

Further investigation into frequencies on a driver level (see App. A, Fig. A.2) revealed a clear divergence in pit stop strategies under SC and VSC conditions. Notably, Lewis Hamilton leads in the frequency of pit stops across both scenarios, with a marked preference for pitting during SC periods. This trend is consistent among most drivers, reinforcing the premise that the longer SC periods offer a strategic pit window that drivers are keen to exploit. The underpinning factors for this preference may include the drivers' track position at the onset of the SC, the state of tire wear, and the calculated risk-reward of pitting relative to maintaining track position.

Moving to the comparison of team tactics on SC/VSC pit stops, the heatmap analysis (see App. A, Fig. A.4) reveals that the strategies are far from homogeneous. Certain teams tend to cluster their pit stops within specific race laps, hinting at predetermined strategies or adaptive responses to unfolding race dynamics. For instance, *Mercedes* tends to conduct pit stops predominantly in the earlier part of the race, a tactic that might reflect a particular approach to tire management or a response to the initial race conditions. In contrast, other teams demonstrate a more evenly distributed pit stop strategy throughout the race, which may indicate a different strategic philosophy or adaptability to the evolving race dynamics.

Further bolstering our findings, additional data visualisations shed light on the position change scores and lap-by-lap dynamics for drivers who pit during SC/VSC periods compared to those who did not. The box plot analysis (see App. A, Fig. A.5) distinctly illustrates the distribution of position changes, where the median of the 'True' group (those who pitted) suggests an advantageous position score gain relative to the 'False' group. However, the wider spread and more significant outliers in the 'True' group suggest a higher variance in outcomes, demonstrating that pitting during SC/VSC periods has the potential for a slight position advancement. Still, it also introduces a greater degree of unpredictability to the results. Building on this understanding, Figure 5's line graph further elaborates on these dynamics. It traces the cumulative position changes of drivers pitted during SC/VSC periods against those who did not, showing clear instances of contrasting outcomes: significant gains for some drivers and losses for others, depending on whether they took a pit stop. Such patterns demonstrate the double-edged nature of pitting decisions, which can offer a competitive edge or lead to a comparative setback, depending on timing and race conditions.

In light of these observations, it becomes evident that the strategic choice to pit during SC/VSC periods is a multifactorial decision with significant consequences. The volatility underscores the

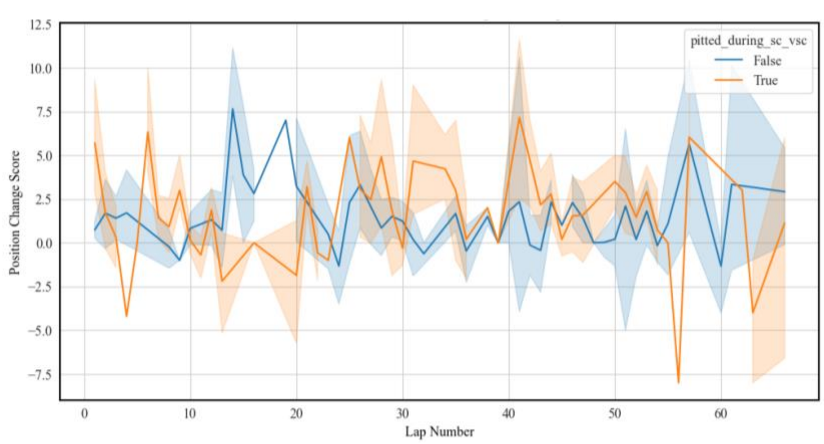

Figure 5: Cumulative Position Change Score across labs

importance of considering a broader spectrum of influencing factors beyond the immediate reaction to SC/VSC deployments.

5.3. Multiple Linear Regression

5.3.1. Primary Regression Model

This part's core methodology involves using multiple linear regression to test the hypothesis that pitting during SC/VSC periods significantly affects a driver's change in its finishing position. This statistical approach quantifies the strength and form of the relationship between multiple independent variables and a single dependent variable, which in this analysis is the *position_change_score*. Multiple linear regression is ideal for testing the hypothesis due to its precision. It estimates the coefficients of the independent variables, which indicate the expected change in the dependent variable for a one-unit change in the predictor, holding other variables constant. This precision enables assessing the specific impact of pitting during SC/VSC periods. To ensure a successful regression analysis that can accurately test the hypothesis, the data must meet certain assumptions (Poole and O'farrell 1970):

- **Linearity**: The relationship between the independent and dependent variables should be linear. *Pittet_during_sc_vsc* is a binary predictor, which implies that there are only two levels and, therefore, no violation of the linearity concern.
- **Independence**: The dataset ensures independence of observations, with each pit stop and non-pit stop lap of a driver treated as a distinct event.
- **Multicollinearity**: The variables' variance inflation factors (VIF) range from 1.08 to 2.92. They are within acceptable limits, which indicates minimal multicollinearity and preserves the integrity of the regression coefficients.
- **Homoscedasticity**: Scatterplot analysis (see App. A, Fig A.6) indicates consistent variance of residuals across fitted values, suggesting homoscedasticity. The clustering of residuals at zero

reflects the discrete nature of racing outcomes, particularly the commonality of no position change during numerous laps.

- **Normality of Residuals**: The Q-Q plot (see App A, Fig A.7) shows an overall linear relationship in the residuals' distribution, affirming their approximate normality. Tail deviations highlight the presence of outliers, which, in the context of racing, represent plausible extremes rather than data issues due to the unpredictability inherent in the sport.

This model leverages these insights to robustly test the hypothesis, focusing on evaluating the strategic impact of pitting during SC/VSC conditions on race outcomes, all while controlling for other influential variables.

5.3.2. Results

The regression analysis aims to unravel the complex dynamics of the pit stop strategy and its impact on the *position_change_score*. The model demonstrates a solid explanatory power, with an R-squared value of 0.369, indicating that nearly 37% of the variability in position change scores can be attributed to the variables included in the model (see App. A, Table A.1). This effectiveness is further substantiated by an adjusted R-squared of 0.329, suggesting a robust fit while accounting for the number of predictors. The model's overall statistical significance is underscored by an F-statistic of 9.051, which, coupled with a p-value of 3.00e-21, decisively indicates that the variables collectively have a meaningful impact on the weighted position changes. A closer look at the residuals' skewness and kurtosis reveals a reasonably normal distribution, albeit with some slight deviations, suggesting a reasonable adherence to the normality assumption. The Durbin-Watson statistic of 1.893 alleviates concerns about autocorrelation in the data, bolstering the reliability of our findings.

Central to the hypothesis testing, the variable *pitted_during_sc_vsc* emerged as a key influencer, exhibiting a statistically significant positive coefficient of 0.3653 ($p = 0.030$), which indicates, on average, pitting during SC/VSC periods is associated with an improvement in the weighted position change score. This improvement is relative to the driver's position immediately following the SC/VSC period. The resulting regression equation, reflecting the interplay of various factors, is as follows:

$$y = 0.3653 * pitted\_during\_sc\_vsc + \beta * \ var_{control} - 0.2148 \quad (2)$$

This formula encapsulates the collective influence of multiple variables, including those related to race strategy and driver performance. Notably, apart from *pitted_during_sc_vsc*, the control variables *overtakes* ($\beta = 0.0786$), and *pos_change_after_sc_vsc* ($\beta = -0.2767$) were also significant, highlighting the importance of driver skill and post-SC/VSC race dynamics. The subtle yet significant role of *all_pitstops_late* ($\beta = -0.0085$) adds an interesting layer, suggesting that strategic decisions made later in the race can have a marginal but discernible impact on outcomes. While some control variables like *tire_change* ($\beta = 0.2731$) and *pit_time_avg_seconds* ($\beta = -0.2215$) did not reach statistical significance, their inclusion is essential for a holistic understanding of the factors influencing race dynamics. Furthermore, the analysis incorporated team-specific variables to explore the influence of team dynamics. Among these, most teams did not show a statistically significant effect on the *position_change_score*. However, the *Haas F1 Team* is a notable exception, which emerged with a significant negative coefficient ($\beta = -0.5497$). This result might suggest that Haas's strategies or other team-specific factors could have influenced their performance. However, while the result could suggest strategic shortcomings, it might also manifest the overall competitive disadvantages faced by Haas compared to other, more successful teams in the field.

In summary, the regression analysis provides valuable insights into the strategic layers of racing, particularly highlighting the crucial role of pitting decisions during SC/VSC conditions. While the model highlights certain influential factors, it's important to note that the explanatory power, as indicated by an R-squared value of 0.369, suggests additional elements at play in determining race outcomes that are not captured within our model.

5.3.3. Cluster-based Regression Model

In our ongoing exploration of racing strategies, particularly focusing on pit stop decisions during SC and VSC periods, we've taken a step further by conducting a multiple linear regression based on the defined driver clusters. The regression results include data for all identified driver clusters (CL0, CL1, CL2, CL3) alongside a comprehensive analysis of the entire driver population (ALL) (see Table 7). Notably, clusters CL0, CL1, and CL3 have smaller sample sizes, which might influence the statistical reliability of our comparisons. Specifically, the higher p-values for F-statistics in CL1 and CL3 limit the statistical significance of their results. Therefore, our focus will primarily be on the largest driver cluster, CL2, which consists of 165 observations and offers a more robust dataset for insightful exploration of how strategic impacts differ among distinct groups of drivers with different characteristics.

In the overall regression (ALL), the variable *pitted_during_sc_vsc* showed a significant positive coefficient, indicating that pitting during SC/VSC periods generally leads to better race outcomes across the driver population. However, when examining CL0 specifically, the coefficient for *pitted_during_sc_vsc* decreases to 0.2602. This reduction suggests that pitting during SC/VSC periods still has a positive impact in CL0, but it's less pronounced compared to the general driver population. This could imply that drivers who focus on strategic manoeuvring and consistency may not leverage SC/VSC pit opportunities to the same extent as other drivers. Furthermore, the significant variables – *overtakes* (β = 0.14) and *all_pitstops_late* (β = -0.0224) – highlight the drivers' proficiency in dynamic racing scenarios and a preference for early or mid-race pit strategies, underscoring a focus on race longevity and tactical positioning.

Shifting our focus to CL2, we see a different tactical approach, emphasising consistency and strategic pit stops over immediate tire wear benefits. The results from this segment yield an R-squared value of 0.439, indicating a reasonable degree of explanatory power within this group. While this marks an improvement from the general model, the adjusted R-squared of 0.362 tempers this view, suggesting that while the model is informative, it doesn't capture the full complexity of factors influencing race outcomes. The F-statistic, at 5.645 with a p-value of 1.49e-10, supports the overall statistical validity of the model for this cluster. Yet, it's essential to approach these results with a degree of caution. Notably, the influence of *pitted_during_sc_vsc* ($\beta = 0.0605$) deviates from both the overall regression and CL0. The reduced coefficient and its loss of significance in this cluster imply that SC/VSC period pitting may not be as advantageous for these drivers, possibly due to their inclination towards early pitting strategies. This variation highlights the distinct responses of different clusters to similar race scenarios. The control variables such as *overtakes*, *position_change_after_sc_vsc*, *pitstop_count_before_sc_vsc*, and *sc_deployed* consistently maintain significance in both the general and segmented analyses, underscoring their impact on race outcomes across clusters.

However, *all_pitstops_late* and *tire_life* lose significance in CL2, indicating a strategic shift towards early pitting and the consistent performance of hard tires. Further analysis reveals that variables such as *tire_change*, *pit_time_avg_seconds*, *sc_count_per_race*, and *driver age* have emerged as significant predictors within CL2. The significance of *tire_change* ($\beta = 0.5824$) and *pit_time_avg_seconds* ($\beta = -0.5122$) particularly underscores the cluster's emphasis on efficient tire management and pit stop execution, aligning with a strategy that values quick and effective response during pit stops. The positive impact of *sc_count_per_race* (β = 0.3506) suggests a strategic adaptation to the frequency of SC deployments, possibly indicating a proactive approach in optimising race strategy under varying conditions. Interestingly, the negative coefficient for *driver_age* (β = -0.5392) indicates that younger drivers within CL2 might be more effectively capitalising on the cluster's specific nuances. This finding suggests a potential edge in adaptability or agility among younger drivers, though it should not detract from the valuable experience and contributions of more seasoned drivers in the cluster. Additionally, the significant coefficient for the *Alpine* (β = 0.8548) team points to a synergistic relationship between the team's strategies and the drivers' performance. This relationship is particularly evident in areas such as tire management and pit stop efficiency, suggesting that the team's approach resonates well with the overall strategic orientation of this cluster.

Table 7: Cluster Regression Results

Variable	**(1)** **ALL** N=330	**(2)** **CL0** N=56	**(3)** **CL1** N=52	**(4)** **CL2** N=165	**(5)** **CL3** N=22
pitted_during_sc_vsc	0.3653*	0.2602	0.4955	0.0605	1.6533
overtakes	0.0786*	0.1400*	-0.0130	0.0606*	0.0454
tyre_change	0.2731	0.5395	-0.2992	0.5824*	3.1186
pos_change_after_sc_vsc	-0.2767*	-0.1323	-0.5272	-0.2576*	-0.0714
all_pitstops_late	-0.0085*	-0.0224*	0.0015	-0.0078	0.0088
pit_time_avg_seconds	-0.2215	-0.1954	-0.3781	-0.5122*	0.8478
sc_count_per_race	0.0554	0.2975	-0.0287	0.3506*	-0.2967
tyre_life	0.0757*	-0.1317	0.0126	0.0994	0.4660
pitstop_count_before_sc_vsc	0.1533*	-0.0504	-0.0479	0.2819*	2.1564
driver_age	2.0726	1.5352	0.0157	-0.5392*	2.9283
SCDeployed	0.3155*	-0.3196	-0.1746	0.5684*	0.6927
...					
R-squared	0.369	0.607	0.422	0.439	0.637
Adj. R-squared	0.329	0.431	0.203	0.362	0.047
F-statistic	9.051	3.446	1.928	5.645	1.081
Prob (F-statistic)	3.00e-21	0.000751	0.0555	1.49e-10	0.473

p values are shown as: * $p < 0.05$

In essence, the segmented regression analysis for CL2 highlights their unique approach to racing, marked by endurance, strategic positioning, and efficient pit and tire strategies. It also underscores the inherent variability and complexity of the sport. It illustrates that strategies effective for one group of drivers may not uniformly apply to all, emphasising the need for tailored strategies that reflect the distinct characteristics and strengths of different driver clusters.

5.3.4. Implications

Our regression analysis testing the impact of pit stops during SC and VSC periods has yielded a pivotal finding: pitting during these periods significantly affects a driver's finishing position. The analysis revealed a positive coefficient for the independent variable *pitted_during_sc_vsc*, indicating that, on average, drivers who pit under these conditions are likely to improve their finishing positions. Consequently, we reject the null hypothesis (H0), which proposed no significant impact of pitting during SC or VSC on a driver's finishing position change. This discovery is a crucial addition to the understanding of pit stop strategy, suggesting that pitting during SC/VSC can offer a strategic advantage under certain circumstances.

From a practical standpoint, these findings have substantial implications for race teams and drivers. The decision to pit during SC/VSC periods emerges as an additional strategic factor that could influence the overall race outcome. However, its effectiveness is not uniform across all situations. The strategy’s impact varies notably among different driver clusters, as seen in our analysis. For instance, the less pronounced significance within CL2 suggests that the benefits of pitting under SC/VSC are inconsistent for drivers. This highlights the necessity for race strategies to be tailored, taking into account the individual driver profiles. Teams and drivers need to weigh this strategy against multiple factors, including the driver's initial position, the team's efficiency in executing pit stops, and the prevailing race conditions, to make informed decisions that optimise performance and capitalise on the strategic opportunities presented by SC/VSC deployments.

Our analysis provides interesting insights into pit stop strategies during SC and VSC periods, though it's essential to consider certain limitations. While the broader seasonal context from 2019 to 2021 is acknowledged in the general analysis, this timeframe still bounds our specific findings. Additionally, we excluded pit stops that occurred during the first lap of SC/VSC periods. This decision was taken to avoid the atypical scenarios often associated with the initial lap, such as early-race incidents. However, this exclusion means that our analysis does not account for the potential impact of these early strategic moves on the overall race outcomes. Furthermore, to manage complexity, we focused on each driver's last SC/VSC pit stop in a race. While this approach allowed us to concentrate on the critical decision-making moments towards the end of the race, it also means that we did not consider the possible cumulative or varied impacts of multiple SC/VSC pit stops occurring in a single race.

These limitations highlight areas for future research. A critical area for further exploration is to investigate the cumulative impact of multiple pit stops within a single race during SC/VSC periods. This exploration would expand on the initial analysis of strategic pitting by examining how a series of pit stop decisions, spread throughout the race, collectively influence a driver's final position. Such a study could reveal interactions between sequential strategic choices and their overall effect on race outcomes, offering valuable insights for refining race tactics during these critical moments.

Common Part II

6. Discussion

In the initial phase of our study, we explored the application of advanced analytics in Formula 1, focusing on how driver clustering might inform pit-stop strategies. Our methodical approach to clustering revealed four distinct driver categories within the dataset. This classification emerged from a comprehensive analysis incorporating three key dimensions: performance, tactical, and behavioural patterns. These dimensions encompassed a variety of metrics, each contributing to a deeper understanding of driver profiles.

Despite the high-performance level uniformly exhibited by the drivers, which led to some similarity in cluster characteristics, our analysis succeeded in distinguishing between nuanced aspects of driver behaviour and strategy. The resulting profiles offered four differentiated interpretations of driver profiles and decision-making styles.

Subsequently, these clusters were instrumental in underpinning various predictive models and statistical analyses related to pit stop strategies. Our investigation spanned several critical areas, including optimal pit timing, tire selection, driver prioritisation, and the influence of safety car periods. Through this analysis, we established associations between the divergent driver profiles and the outcomes of our analytical models. These connections allowed us to attribute specific strategic decisions and outcomes to identifiable cluster characteristics, thereby providing a deeper insight into the interplay between driver behaviour and pit stop strategy.

The research findings underscore the central role of driver clustering in enhancing the efficacy of analytical models within Formula 1. These results from our study imply the critical importance of tailoring analytical models to reflect the distinct characteristics of drivers and teams in Formula 1. This customisation is vital for capturing essential features and characteristics that might otherwise be overlooked in a broader, more generalised approach. Our findings suggest that the success of strategic models in Formula 1 hinges on their ability to account for the unique attributes and behaviours of individual drivers and teams.

This study, while thorough, encounters several limitations that should be considered when interpreting the results. Primarily, the availability of telemetry data through the FastF1 API, limited to the 2019 season onwards, restricts the temporal scope of our analysis of driver clusters and their evolution. Although future seasons will incrementally enrich our dataset, the absence of pre-2019 data constrains our historical analysis.

Additionally, a significant unknown in our study is the detailed information on car characteristics, which likely influences driver performance and tactics and, ultimately, the resulting clusters. The unavailability of these specifics, typically kept confidential by teams, may limit our understanding of the technical factors impacting driver behaviour. We attempted to mitigate this through feature engineering and normalisation techniques but acknowledge that this is an approximation.

Another deliberate exclusion for our initial cluster analysis was races affected by rain. This decision was based on the disproportionate representation of wet races in our dataset, which could potentially skew the results. While this aids in maintaining data consistency, it omits the distinct strategic dynamics and behaviours prevalent in wet conditions. Furthermore, the challenge of quantifying track difficulty, despite having data on corners and marshal lights since 2019, presented another limitation. Our approach to normalising this data was a method to address this issue, but it remains uncertain.

These limitations underscore the need for cautious interpretation of our findings, particularly in their application to strategic decision-making in Formula 1. The clustering methodology applied for the 2019 to 2021 seasons and the resulting application on our advanced analytical models demonstrates promise for future applicability, yet it's important to consider these constraints and assumptions. Despite these challenges, we believe our methodology is robust and adaptable for future seasons.

Taking into account the dynamic nature of motorsport in general and Formula 1 in particular, which is subject to constantly evolving strategies, technologies and rules, our work is not intended to be a final solution but rather food for thought for future work and analysis. As the scope of the data grows over time, there is the potential to deepen and develop our methodology. This promises even deeper and more differentiated insights into the strategic complexity of Formula 1 and motorsports. One key area is the expansion of the dataset. With the FastF1 API continually updating, incorporating data from subsequent seasons would not only improve the robustness of our clustering model but also facilitate longitudinal analyses. Such studies could track the evolution of driver strategies and team tactics over time. If telemetry data for seasons before 2019 becomes accessible, it would offer valuable historical insights, enriching our understanding of the sport's strategic development in response to technological and regulatory shifts.

Rain-affected races, excluded from our current analysis, represent a distinct strategic element in Formula 1. Future research could delve into these scenarios, perhaps through specialised models or by incorporating new features into existing frameworks. This focus could unveil how teams and drivers adapt to the challenges posed by variable weather conditions.

Another promising opportunity is the integration of track characteristics into the analysis. Understanding the influence of different track designs and surface conditions on driver performance and pit stop strategies would add considerable depth to our model. Additionally, while data on specific car builds is largely confidential, future collaborations with Formula 1 teams or utilisation of public data could shed light on the relationship between car technology, driver skills, and strategic choices. Also, the field of data analytics and machine learning is rapidly advancing, offering the potential for refining our clustering model with more sophisticated algorithms. These advancements could handle larger and more complex datasets, providing finer-grained insights into driver performance and pit strategies.

As Formula 1 continues to evolve, so does the opportunity for deeper analytical exploration. The advancements in data collection and analysis present fertile ground for enriching our understanding of this complex sport. By adapting to these changes and refining our methodologies, we can contribute not only to academic discourse but also to the practical strategic toolkit of teams and drivers. This research represents a step towards a more data-driven and detailed comprehension of Formula 1 racing, setting the stage for future scholarly and practical advancements in the field.

7. Conclusion

This thesis has delved into the realm of Formula 1 pit stop strategies, employing advanced analytics to understand how driver clustering informs strategic decisions. We discovered four distinct driver categories based on performance, tactical, and behavioural dimensions, providing a nuanced view of driver profiles within the sport. This classification served as a foundation for examining various facets of pit stop strategy, particularly Safety Car pit stops. The research illustrates the impact of driver characteristics on pit stop strategies in Formula 1.

This study enhances our comprehension of how individual behaviours and decisions intertwine with team strategies by linking divergent driver profiles to strategic outcomes. Using driver clusters in predictive models and statistical analyses has shed light on the complexities of strategic decision-making in this high-speed sport. This thesis contributes to the understanding of Formula 1 racing by offering a data-informed perspective on the strategic elements of the sport. It underscores the value of bespoke strategies that consider the distinct qualities of drivers and teams, highlighting the potential of analytics in refining racing tactics.

In summary, this thesis provides an insightful exploration of the strategic dimensions of Formula 1 pit stops. It offers a clearer picture of how data analytics can be applied to decode the intricacies of racing strategies, enriching our understanding of this dynamic and technologically advanced sport.

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Appendix A: Individual Part

Figures:

- Figure A.1:

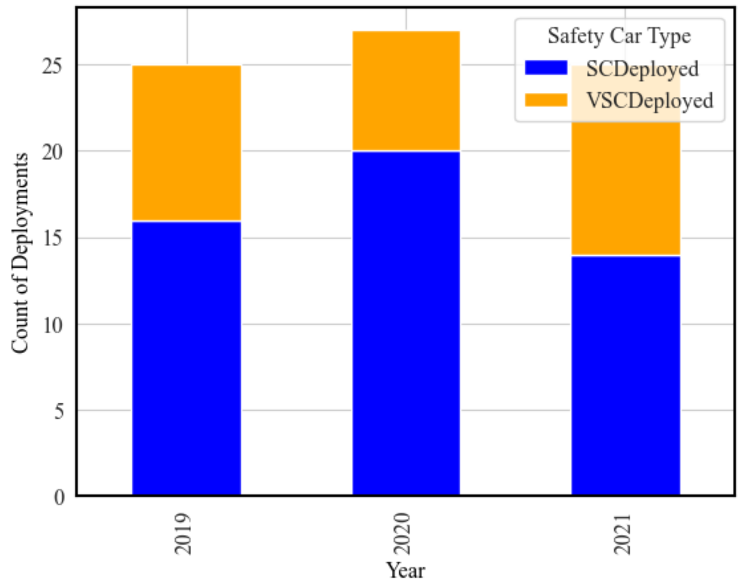

Figure A.1: SC and VSC deployments over seasons

- Figure A.2:

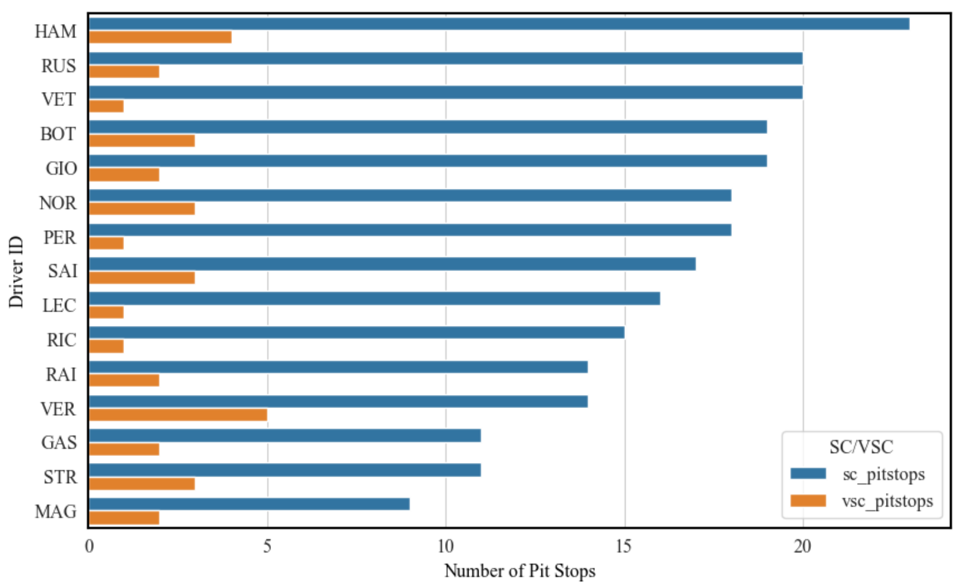

Figure A.2: Comparison of SC and VSC pit stops by drivers

- Figure A.3:

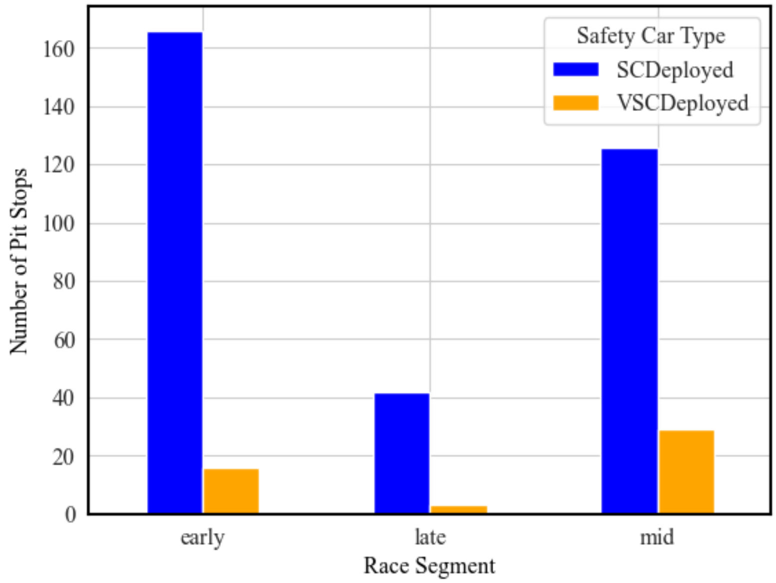

Figure A.3: Distribution of pit stops during SC/VSC across race segments

- Figure A.4:

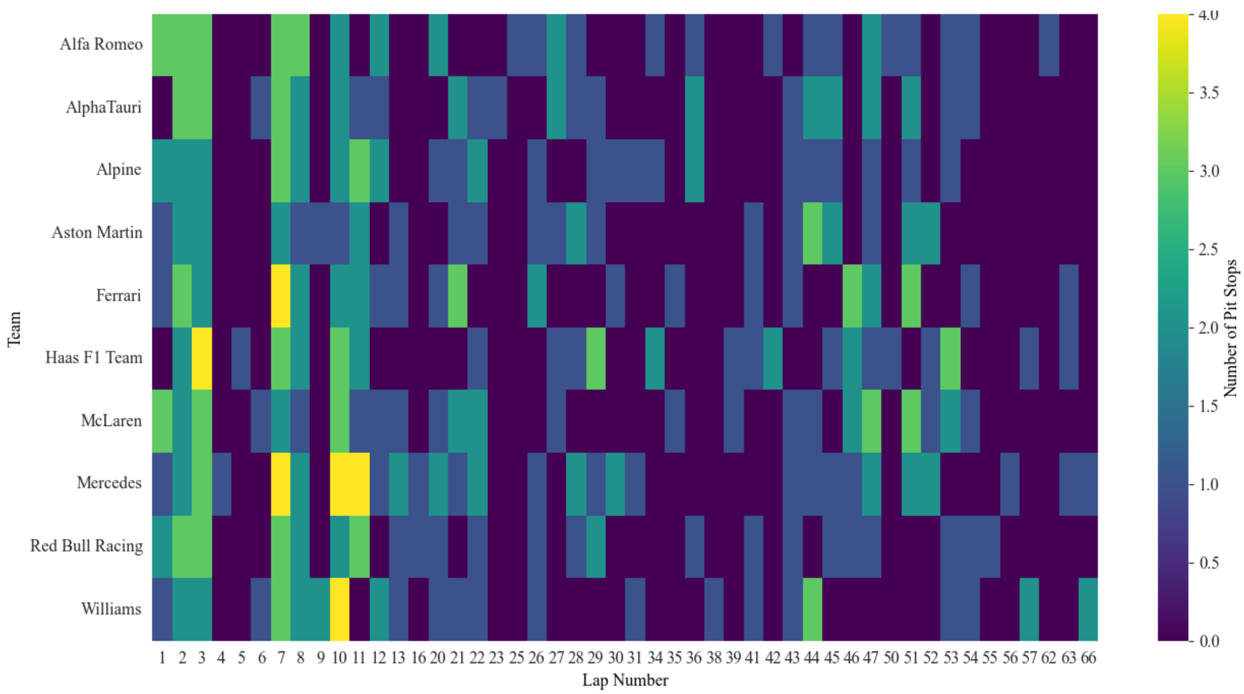

Figure A.4: SC/VSC pit stops across teams and laps

- Figure A.5:

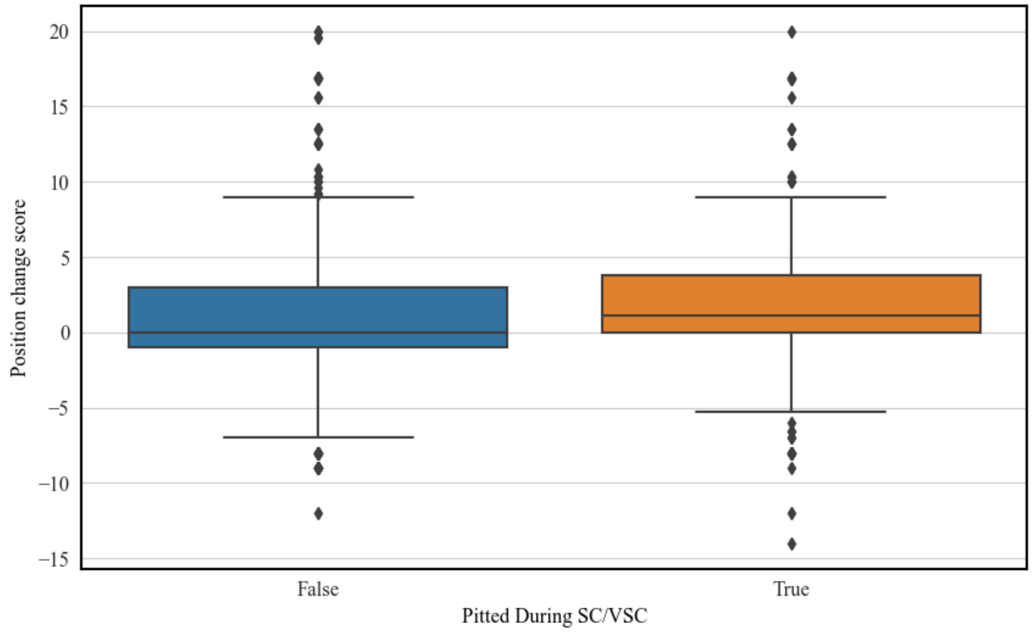

Figure A.5: Comparison of position change score: pitted vs. not pitted during SC/VSC period

- Figure A.6:

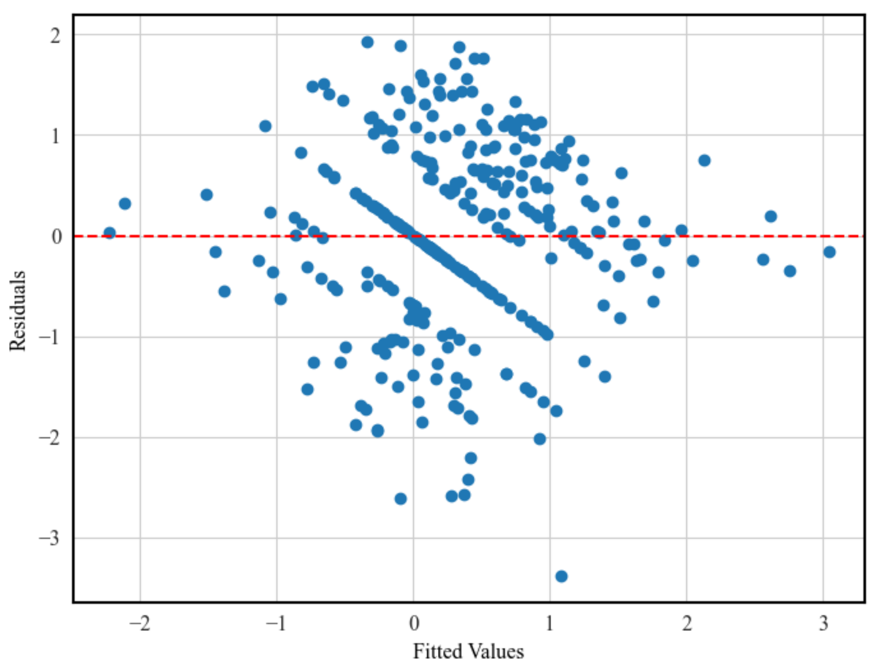

Figure A.6: Scatterplot of the residuals vs fitted values

- Figure A.7:

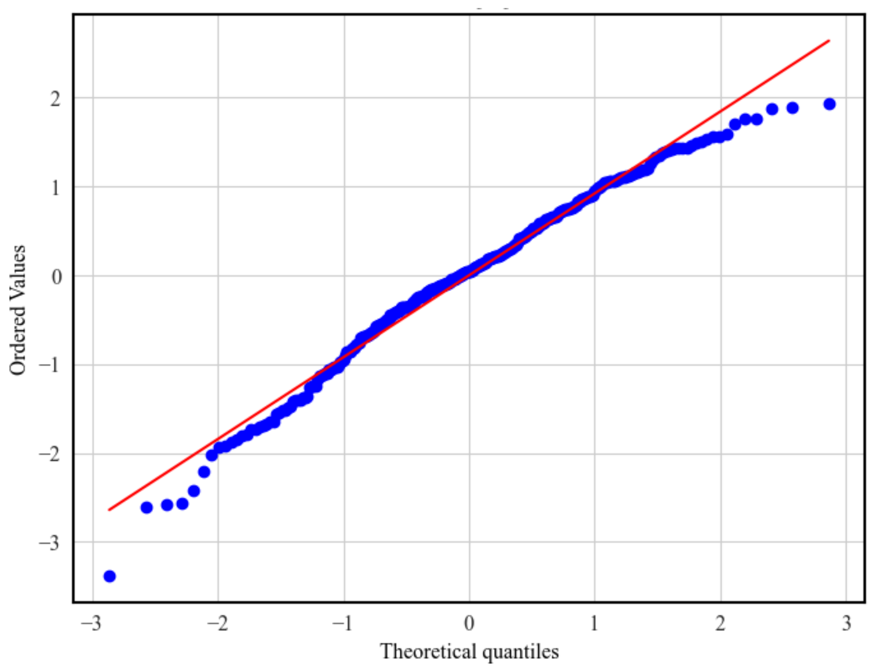

Figure A.7: Q-Q plot of the residuals

Tables:

- Table A.1:

Table A.1: Primary Regression Results

Model Summary				
Dependent Variable	position_change_score		R-squared	0.369
Model	OLS		Adjusted R-squared	0.329
Number of Observations	330		Durbin-Watson	1.893
Df Residuals	309			
Independent Variables	**Coefficient**	**Std. Error**	**t-Statistic**	**p-Value**
const	-0.2148	0.799	-0.269	0.788
pitted_during_sc_vsc	0.3653	0.167	2.186	0.030
Control Variables				
start_position	0.0283	0.016	1.761	0.079
overtakes	0.0786	0.018	4.305	0.000
tyre_change	0.2731	0.180	1.516	0.131
pos_change_after_sc_vsc	-0.2767	0.035	-7.921	0.000
all_pitstops_late	-0.0085	0.003	-2.702	0.007
pit_time_avg_seconds	-0.2215	0.158	-1.400	0.162
tyre_life	0.0757	0.045	1.689	0.092
pitstop_count_before_sc_vsc	0.1533	0.090	1.710	0.088
rainfall	0.2472	0.296	0.836	0.404
SCDeployed	0.3155	0.149	2.110	0.036
Alfa Romeo	-0.1205	0.271	-0.445	0.657
AlphaTauri	-0.2181	0.252	-0.865	0.388
Alpine	0.1441	0.249	0.579	0.563
Aston Martin	-0.4181	0.255	-1.639	0.102
Ferrari	-0.2606	0.234	-1.114	0.266
Haas F1 Team	-0.5497	0.277	-1.983	0.048
McLaren	-0.0049	0.236	-0.021	0.983
Mercedes	0.0794	0.234	0.339	0.735
Williams	-0.2214	0.302	-0.734	0.464

- Table A.2:

Table A.2: Complete Cluster Regression Results

Variable	**(1)** **ALL** N=330	**(2)** **CL0** N=56	**(3)** **CL1** N=52	**(4)** **CL2** N=165	**(5)** **CL3** N=22
const	-0.2148	-6.7719	0.9637	2.6664	-23.1544
pitted_during_sc_vsc	0.3653*	0.2602	0.4955	0.0605	1.6533
start_position	0.0283	0.0209	0.1568	0.0342	-0.0771
overtakes	0.0786*	0.1400*	-0.0130	0.0606*	0.0454
tyre_change	0.2731	0.5395	-0.2992	0.5824*	3.1186
pos_change_after_sc_vsc	-0.2767*	-0.1323	-0.5272	-0.2576*	-0.0714
all_pitstops_late	-0.0085*	-0.0224*	0.0015	-0.0078	0.0088
pit_time_avg_seconds	-0.2215	-0.1954	-0.3781	-0.5122*	0.8478
sc_count_per_race	0.0554	0.2975	-0.0287	0.3506*	-0.2967
tyre_life	0.0757*	-0.1317	0.0126	0.0994	0.4660
pitstop_count_before_sc_vsc	0.1533*	-0.0504	-0.0479	0.2819*	2.1564
rainfall	0.2472	0.1706	-0.2830	0.5928	1.9399
driver_age	2.0726	1.5352	0.0157	-0.5392*	2.9283
SCDeployed	0.3155*	-0.3196	-0.1746	0.5684*	0.6927
Alfa Romeo	-0.1205	-2.6156		0.3637	
AlphaTauri	-0.2181	-0.6346		0.2509	
Alpine	0.1441			0.8548*	
Aston Martin	-0.4181	-1.4764		-0.0115	
Ferrari	-0.2606			0.2268	
Haas F1 Team	-0.5497*	-1.2295		0.3438	
McLaren	-0.0049			0.2823	
Mercedes	0.0794		0.5363		
Williams	-0.2214				
Red Bull Racing		-0.8158	0.4274	0.3555	
R-squared	0.369	0.607	0.422	0.439	0.637
Adj. R-squared	0.329	0.431	0.203	0.362	0.047
F-statistic	9.051	3.446	1.928	5.645	1.081
Prob (F-statistic)	3.00e-21	0.000751	0.0555	1.49e-10	0.473